

DS&A
Experiment:1

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Class/Div: SY3

```
#include<stdio.h>
int length(char[]);
void compare(char[],char[]);
void palindrom(char[]);
void substring(char[], char[]);
void copy(char[],char[]); //Copy second into first
void reverse(char[]);
void displayMenu();
int i,j;
int main(){
    int choice;
    char str[50];
    char str2[50];
    int exit = 0;
    displayMenu();
    do{
        printf("ENTER CHOICE (7 for Display Menu) :\n");
        scanf("%d", &choice);
        if(!(choice==7 || choice==8)){
            printf("Enter String:");
            scanf("%s", &str);
        }
        switch(choice){
        case 1:
            printf("Length of the String is %d\n\n",length(str));
            break;
        case 2:
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        printf("Enter second String: ");
        scanf("%s", &str2);
        copy(str,str2);
        break;

    case 3:
        reverse(str);
        break;
    case 4:
        palindrom(str);
        break;
    case 5:
        printf("Enter substring String:");
        scanf("%s", &str2);
        substring(str,str2);
        break;
    case 6:
        printf("Enter second String:");
        scanf("%s", &str2);
        compare(str, str2);
        break;
    case 7:
        displayMenu();
        break;
    case 8:
        exit = 1;
    }
}while(exit==0);
return 0;
}

void displayMenu(){
    printf("<<-----Menu----->>\n");

```

```
printf("1. Length\n");
printf("2. Copy String in another String\n");
printf("3. Reverse String\n");
printf("4. Check for Palindrom\n");
printf("5. Check for substring\n");
printf("6. compare Any String with another\n");
printf("7. Display menu\n");
printf("8. Exit\n");
}
```

```
int length(char a[]){
    int len = 0;
    //TO DO
    while(a[len] != '\0'){
        len++;
    }
    return len;
}

void compare(char str[50], char str2[50]){
    int flag = 0;
    i=0;
    // TO DO
    while(str[i] != '\0' && str2[i] != '\0'){
        if(str[i] != str2[i]){
            flag = 1;
            break;
        }
        i++;
    }
    if(str[i] != '\0' || str2[i] != '\0'){
        flag = 1;
    }
}
```

```

    if(flag==0){
        printf("Both Strings are Equal\n\n");
    }
    else{
        printf("Both Strings are Not Equal\n\n");
    }
}

void palindrom(char a[]){
    int l = length(a);
    int flag = 0;
    // TODO
    for(i=0,j=l-1;i<j;i++,j--){
        if(a[i]!=a[j]){
            flag=1;
            break;
        }
    }
    if(flag==0){
        printf("The String is Palindrome\n\n");
    }
    else{
        printf("The String is Not Palindrome\n\n");
    }
}

void substring(char a[50], char b[50]){
    // b is substring of a
    int al = length(a);
    int bl = length(b);
    int flag = 0;
    int count = 0;
    //TODO
    if(bl<=al){

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for(i=0;i<=al-bl;i++){
    count=0;
    for(j=0;j<bl;j++){
        if(a[i+j]==b[j]){
            count++;
        }
        else{
            break;
        }
    }
    if(count==bl){
        flag++;
    }
}
if(flag==0){
    printf("The String B is not substring of A\n\n");
}else{
    printf("The String B is substring of A and it occur %d
times\n\n",flag);
}

```

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} else {
    printf("The String B Is not subString of A\n\n");
}
}

```

```

void copy(char a[], char b[]){
    int al = length(a);
    int bl = length(b);
    char str3[50];
    //TODO
    for(i=0;i<bl;i++){
        str3[i]=b[i]; }
}

```

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    str3[bl]='\0';
    printf("Copied String : %s\n\n",str3);
}
void reverse(char a[]){
    int al = length(a);
    //TODO
    char temp;
    for(i=0,j=al-1;i<j;i++,j--){
        temp=a[i];
        a[i]=a[j];
        a[j]=temp;
    }
    printf("Reversed String : %s\n\n",a);
}

```

Output:

Output	Output
<pre> <<-----Menu----->> 1. Length 2. Copy String in another String 3. Reverse String 4. Check for Palindrom 5. Check for substring 6. compare Any String with another 7. Display menu 8. Exit ENTER CHOICE (7 for Display Menu) : 1 Enter String:urvi Length of the String is 4 ENTER CHOICE (7 for Display Menu) : </pre>	<pre> <<-----Menu----->> 1. Length 2. Copy String in another String 3. Reverse String 4. Check for Palindrom 5. Check for substring 6. compare Any String with another 7. Display menu 8. Exit ENTER CHOICE (7 for Display Menu) : 2 Enter String:Urvi Enter second String: Shivalkar Copied String : Shivalkar ENTER CHOICE (7 for Display Menu) : </pre>

```

Output
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrom
5. Check for substring
6. compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
3
Enter String:Urvi
Reversed String : ivrU

ENTER CHOICE (7 for Display Menu) :

```

```

Output
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrom
5. Check for substring
6. compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
4
Enter String:madam
The String is Palindrome

ENTER CHOICE (7 for Display Menu) :

```

```

Output
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrom
5. Check for substring
6. compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
5
Enter String:addition
Enter substring String:add
The String B is substring of A and it occur 1 times

ENTER CHOICE (7 for Display Menu) :

```

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrom
5. Check for substring
6. compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
6
Enter String:urvi
Enter second String:urvi
Both Strings are Equal

ENTER CHOICE (7 for Display Menu) :
```

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrom
5. Check for substring
6. compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
7
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrom
5. Check for substring
6. compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
```

Output

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrom
5. Check for substring
6. compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
8
```

=== Code Execution Successful ===

